

Loïc Magnan

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EDUCATION

University of Illinois Urbana-Champaign

Expected May 2027

- Bachelor of Science in Nuclear Engineering, Concentration in Power, Safety, and Environment (Fission)
- GPA: 3.91, Dean's List
- Relevant Coursework: Applied Large Language Models, Numerical Methods for Neutron Transport & Monte Carlo, Neutron Diffusion, Thermal Hydraulics, Fluid Mechanics, Statistics and Probability
- American Nuclear Society (ANS) - Active Member

EXPERIENCE

Subcritical Systems | AI Engineering Intern

May 2026 – Present

- Built an intelligence layer supporting engineers designing an accelerator-driven subcritical (ADS) power plant.
- Architected a multi-agent system: specialized agents under a central orchestrator, hybrid graph RAG retrieval, provenance controls tracing every output to the source design report, and MCP connections to analysis tools.
- Leveraged this knowledge layer to fill gaps in the design and generate full-facility CAD renderings of the power plant.
- Conducted linear particle accelerator radiation shielding studies in Geant4 with the generated geometry, enabling rapid iteration on shielding design.
- Designed the end-to-end pipeline the company now uses in production to iterate on design decisions and author outgoing EPC and NRC licensing documents; handed off operation to a team of 5 engineers.

Analysis of Reactor Transients and Stability (ARTS) | Research Assistant

Sep. 2024 – Present

- Undergraduate researcher in Prof. Kozlowski's group, which consults for startups on reactor design and feasibility.
- Modeled burnup in subcritical systems with different particle beams and spent fuel compositions.
- Developed a full life cycle techno-economic analysis (TEA) for the ARPA-E subcritical system; currently applying the model to compare data center and flexible-load operation profitability.
- Contributing to the development of the High Temperature Incandescent Reactor, which uses nuclear thermal propulsion (NTP) fuel materials and no coolant to generate power.
- Building from-scratch GPT-2 small pretrains using OpenWebText and gradient to evaluate how neural-network architecture effects model performance.

Center for Plasma-Material Interactions | Lab Assistant

Oct. 2023 – Sep. 2024

- Studied the deposition of stannane gas (SnH₄) on novel materials in high vacuum for ASML lithography applications.

PUBLICATIONS & CONFERENCE PRESENTATIONS

Nuclear Data Needs and Neutronics Analysis of XDS

Apr. 2026

- Wen, R., Fang, M., Magnan, L., Kozlowski, T., Di Fulvio, A. Poster presented at the ARPA-E Energy Innovation Summit (ARPA-E NEWTON project on transmutation of used nuclear fuel).
- Implemented and optimized the accelerator beam using CERN's FLUKA, and performed the neutronics and burnup calculations in Serpent.

Regulatory Feasibility of Contemporary Atmospheric NTP Testing

May 2026

- Erkiralp, D.C., Magnan, L., Kozlowski, T. Accepted to ANS NETS 2026; also presented at STARFIRE 2026.
- Contributed Serpent neutronics and burnup modeling of a ZrC-coated UC2 prismatic NTP core.
- Quantified the effects of fission product release and cladding erosion using EPA CAP-88 atmospheric dispersion analysis for 40 CFR 61 Subpart H compliance.

Bitcoin Mining as a Potential Enabler for Low Cost Fully Nuclear Energy Grids

Apr. 2026

- Magnan, L. (sole author). Presented at the ANS 2026 Student Conference.
- Developed a techno-economic analysis of an intermittent, interruptible mining load on the French grid.
- Explored flexible off-take as an enabler for fully nuclear microgrids using small modular reactors.

ADDITIONAL INFORMATION

AI Engineering: Multi-agent Systems, Claude Agent SDK, MCP, Hybrid graph RAG, Automated CAD (Blender), Jev

Nuclear Codes: Serpent, OpenMC, FLUKA, Geant4, EPA CAP-88

Languages: English, French, Fluent Spanish

Interests: Golf, Squash, Tennis, Local AI, Cooking

Legal Status: F1 Visa holder